Agency and Learning Efficiency as the Structural Invariants of Intelligence

Patrick D. McCarthy

Abstract

We prove that for any entity persisting under bounded active context, survival pressure, and positive entropy: (1) agency $\gamma > 0$ is necessary — the lineage argument shows that $\gamma = 0$ lineages contract and are eliminated; (2) learning efficiency $\eta_M > 0$ is necessary — without it, the active context L_n fills without relief and inference degrades to failure; (3) the action space A (informed actions) and the higher-order function space M (learning) are type-irreducible — no learning process changes a function's type signature; (4) every other property of the entity can equal zero for some persistent entity; (5) therefore $I(E) = \gamma \cdot \eta_M$ is the unique measure of intelligence, derived not stipulated. This completes the structural theory: from bounded context alone, the complete architecture of intelligence follows as necessary consequences.

1 Introduction

This paper completes a chain of results deriving the architecture of intelligence from a single generative constraint: bounded active context.

Paper 1 [13] established that any entity accumulating propositions under bounded context necessarily maintains a directed graph with typed edges. Paper 2 [14] established that bounded context under growing action count produces escalation dominance over top-down allocation. Paper 3 [15] established that survival pressure makes knowledge and action inseparable, with an asymmetric retention criterion. Paper 4 [16] established that agency with feedback from the world produces gradient descent on uncertainty, with convergence, and that cost-of-failure generates an encoding hierarchy with permanence.

One question remains: **what properties must every persisting entity have?** We prove that the answer is exactly two: agency $\gamma > 0$ and learning efficiency $\eta_M > 0$. Their product is intelligence.

This result engages a long-standing difficulty. Dupoux, LeCun, and Malik [6] deliberately avoid defining intelligence, instead characterizing it as the capacity for autonomous learning. We derive the definition they avoided giving. Hutter [11] and Legg and Hutter [12] define intelligence as average performance over computable environments; we define it as rate of improvement, independent of current performance level. Chollet [4] measures skill-acquisition efficiency — structurally close to η_M but treated as a benchmark metric rather than a structural invariant. Schmidhuber [19] identifies curiosity as compression progress, which corresponds to η_M, but treats drive as a design objective rather than a necessary condition of persistence. Friston's free energy principle [9] derives minimization behavior from self-organization but does not establish the minimization *imperative* — why $\gamma > 0$ is structurally necessary.

The contribution of this paper is to prove, via a selection argument on lineages, that $\gamma > 0$ and $\eta_M > 0$ are the unique structural invariants of persistence, and that $I(E) = \gamma \cdot \eta_M$ is the only consistent intelligence measure.

2 Formal Framework

We build on the definitions of Papers 1–4 and introduce the structures needed for the present results. Throughout, W is a measurable space of world states with positive entropy, K is a knowledge base of propositions about W, C_n is the bounded active context capacity, A is the action space (actions informed by knowledge), and $U(w, K) = -\log P(w \mid K)$ is the uncertainty of K about state w [21, 5].

Definition 1 (Higher-Order Function Space M)**.** Let $A : K \times W_{\text{obs}} \to \Delta\mathcal{A}$ be the action space of informed actions, where W_{obs} denotes observable world states and $\Delta\mathcal{A}$ the simplex over available actions. The *higher-order function space* is

$$M : (A, K) \times L \;\to\; (A, K)$$

where L is a learning signal (feedback from W following action). M is categorically different from A: elements of A map knowledge and observations to action distributions; elements of M map action–knowledge pairs and learning signals to updated action–knowledge pairs.

M is the cross product of all dimensions not yet precipitated into concrete knowledge or action:

$$M = \prod_i D_i$$

where $\{D_i\}$ are the latent dimensions of the entity's adaptive capacity. As the entity scales, dimensions precipitate from M into K or A — the same normalization argument by which knowledge precipitates from action [15].

The hierarchy is open upward: $M^{(2)}$ operates on $M^{(1)}$, yielding meta-learning; $M^{(n+1)}$ operates on $M^{(n)}$. No natural termination point is established (see Section 8).

Definition 2 (M Efficiency)**.** Let $G(a, K) = I(a; W \mid K)$ denote the mutual information between action a and the true world state W, conditional on knowledge K — the *grounding* of a. Given a learning event where $(A', K') = M((A, K), L)$, define the *learning efficiency*:

$$\eta_M \;=\; \mathbb{E}_L\left[\frac{\max_{a' \in A'} G(a', K') - \max_{a \in A} G(a, K)}{|L|}\right]$$

Both numerator and denominator are measured in bits; η_M is dimensionless. The three regimes:

- $\eta_M > 0$: the entity learns — grounding improves per unit of learning signal.
- $\eta_M = 0$: the entity executes without learning — actions are informed by fixed (A, K) but never updated.
- $\eta_M < 0$ temporarily: exploration, creative destruction — locally irrational steps that may increase long-run survival. Persistently $\eta_M < 0$: the entity does not persist.

Remark 1 (Measurement). Computing η_M requires evaluating $G(a, K) = I(a; W \mid K)$, which depends on the true joint distribution $P(a, W, K)$. The entity cannot compute its own η_M — it would need the very information it is trying to learn. This is a structural feature, not a defect: self-knowledge of learning rate would require solving the problem that learning is meant to solve. Both γ and η_M are measurable by an external observer with access to the entity's trajectory and outcomes.

Definition 3 (Intelligence)**.** For an entity E with agency γ and learning efficiency η_M:

$$I(E) = \gamma \cdot \eta_M$$

This is stated here as a definition. Theorem 3 establishes that it is the unique measure consistent with the structural invariants of persistence.

Definition 4 (Agency)**.** Agency $\gamma \in [0, 1]$ is the probability that the entity engages its adaptive capacity M in response to a learning signal. Equivalently, γ is the response gate: the fraction of environmental signals that trigger an update to (A, K).

Operationally:

$$\gamma \propto \lim_{U_{\text{lethal}} \to \infty} \frac{dU}{dt}$$

As the lethality of the environment increases without bound, an entity with $\gamma > 0$ increases its rate of uncertainty reduction; an entity with $\gamma = 0$ does not respond.

3 Type Irreducibility of A and M

Theorem 1 (Type Irreducibility of A and M)**.** *No element of A can become an element of M through any learning process. No element of M can become an element of A through any learning process. The two spaces are type-irreducible.*

Proof. The type signatures are distinct:

$$A : K \times W_{\text{obs}} \to \Delta\mathcal{A}$$
$$M : (A, K) \times L \to (A, K)$$

A takes knowledge and observations as input and returns an action distribution. M takes an action–knowledge pair and a learning signal as input and returns an updated action–knowledge pair. The domains and codomains are different types.

No learning process changes a function's type signature. An element $a \in A$ that achieves certainty (its output concentrates on a single action for all inputs) promotes to a lower encoding level in the hierarchy established in [16] — but this is a change of encoding level, not a change of type. The promoted a still maps $K \times W_{\text{obs}} \to \Delta\mathcal{A}$; it is still an element of A.

Conversely, M takes (A, K) as input. An element of A does not accept (A, K) as input — it accepts $K \times W_{\text{obs}}$. No amount of learning can change what a function accepts as input. The type distinction is categorical, not parametric.

Why unification fails. A natural objection holds that the type distinction is definitional: we defined A and M to have different signatures, so of course they are different. The non-trivial content is that any entity attempting to collapse A and M into a single type T faces an impossibility. Suppose $T : K \times W_{\text{obs}} \times (T, K) \times L \to \Delta\mathcal{A} \times (T, K)$. An element of T must simultaneously accept observations (to select actions) and accept its own space as input (to modify itself). But self-application $T(T)$ is ill-typed in any consistent type theory [2]: it produces the Russell-style paradox of a function that modifies itself. Typed lambda calculi resolve this by stratification — exactly the A/M separation. The irreducibility is not our assumption but a consequence of type-theoretic consistency. □

Corollary 2 (Distinct Roles)**.** *The type irreducibility establishes distinct functional roles:*

- *A navigates W given K: selecting actions in the current world state using current knowledge.*
- *M evolves (A, K): modifying the entity's action repertoire and knowledge base in response to learning signals.*

Concrete instances of M acting on different targets:

- ***Neuroplasticity**: M acting on A — modifying which actions are available and how they respond to inputs.*
- ***Insight**: M acting on K — restructuring the knowledge base to incorporate a new proposition that changes the graph topology.*
- ***Habit formation**: M promoting from L_n to lower encoding levels — the process described in [16], Theorem 4.*
- ***Evolution**: M acting on M itself — the highest level of the hierarchy, operating at the most stringent rigor threshold.*

4 Necessity of γ and η_M

Theorem 3 (Structural Invariants of Persistence)**.** *For any entity E persisting under bounded active context C_n, survival threshold U^d_{lethal}, in a world W with positive entropy:*

(i) $\gamma(E) > 0$ *necessarily.*

(ii) $\eta_M(E) > 0$ *necessarily (in expectation over the entity's trajectory).*

(iii) Every other scalar property of E can equal zero for some persistent entity.

(iv) Therefore $I(E) = \gamma \cdot \eta_M$ is the natural measure of intelligence — identified by what survival requires.

Scope note. This theorem establishes $\gamma > 0$ and $\eta_M > 0$ as *necessary* conditions for persistence. The proof is a selection argument: entities lacking either property are eliminated. The theorem does *not* explain the *origin* of γ — how $\gamma > 0$ structures bootstrap from $\gamma = 0$ substrates is an open question (see Section 8).

Scope clarification. The theorem applies to entities with the structure $E = (K, A, \sigma, \gamma)$: a knowledge base, an action space, a sensing function, and agency. Inert objects (rocks, tables) lack this structure — they have no K, no A, no σ — and their persistence is not governed by the theorem. The theorem's domain is entities that *face* survival thresholds U^d_{lethal} because they *have* uncertainty about world states. An object without a knowledge base has no uncertainty and no survival threshold in the relevant sense.

Proof. **Part 1: Necessity of $\gamma > 0$ — The Lineage Argument.**

Suppose $\gamma(E) = 0$. By Definition 4, the entity never engages M: its adaptive capacity is zero. Therefore $(A(t), K(t)) = (A(0), K(0))$ for all t — the entity's action–knowledge pair is fixed at its initial state.

W has positive entropy. This is a thermodynamic fact, not a modeling assumption: any world with energy gradients produces states not contained in any finite knowledge base $K(0)$.

Formalization. "Positive entropy" is used here in the sense of positive entropy rate: $H'(W) > 0$, meaning the environment generates new information per unit time [5]. This is weaker than requiring W to be high-dimensional — a one-dimensional chaotic system has positive entropy rate. It excludes only trivially static or periodic environments. The connection to persistence is a consequence of Ashby's law of requisite variety [1]: an entity's adaptive capacity must match the variety of the environment, and positive entropy rate ensures that variety grows without bound. A $\gamma = 0$ entity has zero adaptive capacity; its requisite variety deficit grows monotonically.

The selection argument operates on *lineages*, not individuals. An entity with $\gamma = 0$ reproduces copies that inherit the same fixed $(A(0), K(0))$. Copies disperse across W. As the lineage's collective coverage of W grows with each generation, even though each individual's (A, K) is static, the lineage encounters novel configurations of W where

$$U(w, K(0)) > U^d_{\text{lethal}}$$

No copy can adapt ($\gamma = 0$); copies in those regions of W are removed. Copies in locally stable niches persist temporarily, but niches are not permanent: W has positive entropy, and no region of the state space remains static indefinitely. The $\gamma = 0$ lineage contracts until eliminated. Selection removes the lineage.

The enforcement operates at multiple timescales:

- **Individual**: $\gamma = 0$ entities in novel environments are directly removed when $U(w, K(0)) > U^d_{\text{lethal}}$.
- **Evolutionary**: lineages without $\gamma > 0$ offspring are eliminated as their collective niche erodes.
- **Cultural**: groups without institutional agency ($\gamma > 0$ at the organizational level) are outcompeted by groups that learn.

Therefore $\gamma(E) > 0$ is necessary for persistence.

Part 2: Necessity of $\eta_M > 0$.

Suppose $\gamma > 0$ but $\eta_M \leq 0$.

Case $\eta_M = 0$: The entity has drive ($\gamma > 0$) but M produces no improvement in grounding. (A, K) is effectively fixed despite engagement. This reduces to the situation of Part 1: the entity's adaptive capacity produces no adaptation, so the lineage argument applies identically.

Additionally, C_n is bounded. Novel world states generate learning signals that fill the active context L_n with unresolved content. By the escalation result [14], inference quality ρ degrades as L_n fills without promotion to lower encoding levels. Promotion requires $\eta_M > 0$: achieving the rigor threshold θ_i at encoding level i requires M to improve the relevant action to the required certainty (Theorem 4 of [16]). With $\eta_M = 0$, no action reaches threshold, L_n is never relieved, and $\rho \to 0$. Inference fails.

Case $\eta_M < 0$ *persistently*: (A, K) worsens in expectation with each learning event. Grounding decreases; uncertainty increases. This accelerates the failure mode of the $\eta_M = 0$ case.

The persistence condition is:

$$\mathbb{E}_t[\eta_M] > 0 \quad \text{over the entity's trajectory}$$

not $\eta_M(t) > 0$ at every time step. Temporary $\eta_M < 0$ (exploration, creative destruction) is compatible with persistence and can increase long-run survival. What is required is that the trajectory-averaged learning efficiency is positive.

Therefore $\mathbb{E}_t[\eta_M] > 0$ is necessary for persistence.

Part 3: Minimality — every other property can equal zero.

For every other scalar property of E, there exists a persistent entity for which that property equals zero:

- $K(0) = \varnothing$: an entity with $\gamma > 0$ and $\eta_M > 0$ survives starting from empty knowledge — it builds K from gradient descent on U [16].
- $A(0) = \{\text{id}\}$: the identity action still returns feedback from W, grounding the first proposition.
- $C_n = C_{\min}$: minimal context capacity. Graph structure [13] maintains $\rho = 1$ by retrieving exactly the relevant subgraph per problem.
- Substrate, sensory bandwidth σ, observable window W_{obs}: these are free parameters. No specific value is required for persistence; they constrain the rate and form of learning but not its possibility.

No other scalar property has the survival-critical necessity established in Parts 1 and 2.

Part 4: $I(E) = \gamma \cdot \eta_M$ follows by arithmetic.

Parts 1–3 are the substantive content: γ and η_M are the unique structural invariants of persistence. Given that γ is a probability and η_M a conditional rate, their product is the expected improvement rate by the law of total expectation (Corollary 4). The discovery is in Parts 1–3; the product form is a consequence of what the invariants *are*, not a separate result. □

5 $I(E) = \gamma \cdot \eta_M$ Is Structurally Necessary

Corollary 4 (Uniqueness of the Intelligence Measure)**.** *γ and η_M are the unique structural invariants of persistence (Theorem 3). Any intelligence measure must be:*

- *Positive if and only if both $\gamma > 0$ and $\eta_M > 0$.*
- *Zero if either $\gamma = 0$ or $\eta_M = 0$.*

The product $I(E) = \gamma \cdot \eta_M$ is the unique normalized combination satisfying these constraints.

Proof. $\gamma \in [0, 1]$ is a probability (the response gate). η_M is a conditional rate (bits of grounding improvement per bit of learning signal, conditional on engagement). By the law of total expectation:

$$\mathbb{E}[\text{improvement rate}] = \gamma \cdot \eta_M + (1 - \gamma) \cdot 0 = \gamma \cdot \eta_M$$

The entity either engages M (with probability γ, yielding improvement rate η_M) or does not engage (with probability $1-\gamma$, yielding zero improvement). The expected improvement rate is the product. This is not a choice of combination rule; it is the law of total expectation applied to a probability and a conditional rate.

Alternative combinations fail:

- $\gamma + \eta_M$: assigns positive intelligence to an entity with $\gamma = 0$ and $\eta_M > 0$ (a zero-drive entity that would learn if it engaged but never engages). This contradicts Part 1 of Theorem 3.
- $\min(\gamma, \eta_M)$: erases the independent variation of the two invariants. An entity with $\gamma = 0.01$ and $\eta_M = 100$ (rare engagement, high efficiency) and an entity with $\gamma = 0.01$ and $\eta_M = 0.01$ (rare engagement, low efficiency) receive the same score despite vastly different persistence prospects.
- Any function $g(\gamma, \eta_M)$ satisfying the constraints of Corollary 4 must be zero on the axes and positive in the interior. Among functions of a probability and a conditional rate, the product is the unique form given by the law of total expectation.

The product $\gamma \cdot \eta_M$ has the same structure as the retention criterion of Theorem 1 in [15]: probability of needing it $\times$ conditional value given need. This is not a coincidence — both are instances of expected value under a binary gate. □

Remark 2 (Exploration and Creative Destruction). Temporarily negative η_M — locally irrational steps that reduce grounding in the short run — is survival-optimal when the persistence condition requires only $\mathbb{E}_t[\eta_M] > 0$ over the trajectory, not $\eta_M \geq 0$ at every step. Exploration (trying actions with uncertain outcomes) and creative destruction (dismantling existing (A, K) structure to rebuild it better) are the signature of M sophisticated enough to optimize over trajectories rather than individual steps. An entity restricted to $\eta_M \geq 0$ at every step is trapped in local optima; an entity that can tolerate temporary $\eta_M < 0$ accesses a larger region of the (A, K) landscape.

6 Engagement with Prior Work

Dupoux, LeCun, and Malik [6]. They deliberately avoid defining intelligence, instead characterizing it as the capacity for autonomous learning. Theorem 3 establishes that this is structurally correct: $\gamma > 0$ (autonomous engagement) and $\eta_M > 0$ (learning efficiency) are the unique invariants. Their product is the definition they avoided giving. Their dual-system architecture (observation + action with meta-control routing) corresponds to the A/M distinction established in Theorem 1, with meta-control corresponding to $M^{(2)}$.

Hutter [11] and Legg–Hutter [12]. They define intelligence as average performance over computable environments, formalized via AIXI. Three structural differences from the present account: (1) AIXI assumes unbounded computation; we have bounded C_n. (2) They define intelligence as performance level; we define it as rate of improvement (η_M), independent of current performance. A novice chess player learning rapidly has higher η_M than a grandmaster at plateau, despite lower performance. (3) Their environment class is all computable functions; ours is physical W with positive entropy — a narrower but physically grounded class.

Chollet [4]. Skill-acquisition efficiency, the core of the Abstraction and Reasoning Corpus, is structurally close to η_M. Two differences: Chollet treats prior knowledge K as a confound to be controlled for in measurement; here K is constitutive, because η_M operates on (A, K) and the efficiency of learning depends on what is already known. Chollet's "scope" and "generalization difficulty" map to the growth of the reachable territory $T_{\text{reachable}}$ in the (A, K) landscape.

Schmidhuber [19]. Curiosity as compression progress corresponds precisely to η_M: the agent seeks states that maximize improvement in its world model. The difference is foundational. Schmidhuber treats intrinsic drive as a design objective — a reward signal to be engineered into artificial

agents. Theorem 3, Part 1, establishes $\gamma > 0$ as a necessary condition of persistence: not a reward to be designed in, but a property that selection enforces.

Friston [9, 10]. The free energy principle derives minimization behavior from self-organization: systems that persist must minimize surprise (free energy). This is consistent with gradient descent on U [16]. The gap is the minimization *imperative*: FEP establishes that self-organizing systems *do* minimize free energy, but does not establish why $\gamma > 0$ is structurally necessary — why entities that fail to minimize are eliminated. Theorem 3, Part 1, fills this gap via the lineage argument: $\gamma = 0$ lineages contract under positive entropy regardless of their initial fitness.

Cattell [3] and Spearman [23]: Psychometric intelligence. Cattell's distinction between fluid intelligence (G_f) and crystallized intelligence (G_c) maps onto the framework. G_f — the capacity for novel problem-solving independent of prior knowledge — is structurally close to η_M: the rate at which the entity extracts grounding from new learning signals. G_c — accumulated knowledge and skill — is $|K|$, the size of the knowledge base. The framework predicts the well-established correlation between G_f and G_c: entities with higher η_M build larger K faster, so the two measures co-vary despite measuring different quantities.

Spearman's g-factor — the dominant factor across psychometric tests — should, on this account, correlate with $I(E) = \gamma \cdot \eta_M$. This is a testable prediction: g should load on both engagement rate (willingness to attempt novel problems) and learning efficiency (improvement per attempt), and the product should predict g better than either factor alone.

7 The Complete Chain

The five papers derive the complete architecture of intelligence from a single generative constraint. Each step is a consequence of bounded active context interacting with properties of the physical world:

1. **Bounded C_n + information preservation $\Rightarrow$ graph structure** [13]. Any entity accumulating propositions under bounded context necessarily maintains a directed graph with typed edges.

2. **Bounded C_n + growing A $\Rightarrow$ escalation dominates top-down** [14]. As the action count grows, escalation pressure from the top of the encoding hierarchy dominates, forcing reorganization.

3. **Survival pressure $\Rightarrow$ K/A inseparability with asymmetric retention** [15]. Under survival pressure, knowledge and action become inseparable; the retention criterion is probability of need $\times$ conditional value.

4. **$\gamma > 0$ + feedback from W $\Rightarrow$ gradient descent on U with convergence; cost-of-failure $\Rightarrow$ encoding hierarchy** [16]. Agency with feedback produces convergent uncertainty reduction. Asymmetric costs generate a permanent encoding hierarchy.

5. **Positive entropy + bounded C_n $\Rightarrow$ $\gamma > 0$ and $\eta_M > 0$ necessary; $I(E) = \gamma \cdot \eta_M$ unique measure** [this paper]. The lineage argument and the context-filling argument establish the two invariants. Their product is the unique intelligence measure.

Each step is a consequence of the same generative constraint. The contribution is not any individual result but the connected chain: from bounded context to graph structure to survival-driven retention to escalation to gradient descent to the structural definition of intelligence.

8 Open Questions

1. **Origin of γ.** Theorem 3 establishes that $\gamma > 0$ is necessary for persistence but does not explain how $\gamma > 0$ structures bootstrap from substrates where $\gamma = 0$. The thermodynamic conjecture: dissipative structures [18, 7] under sustained energy flow spontaneously produce $\gamma > 0$ configurations, which selection then retains. This is distinct from the necessity result.

2. **Direction of γ.** Is agency uniformly distributed across learning signals, or does it have intrinsic directedness? If directed, what determines the direction?

3. **The $M^{(n)}$ hierarchy.** How many levels of meta-learning does a persistent entity require? Is there a natural termination point, or is the hierarchy open upward without bound?

4. **Bound composition in A.** How do individual action bounds compose when multiple actions share knowledge substructure?

9 Conclusion

Intelligence is the driven capacity to improve. This is not proposed but derived. The lineage argument (Theorem 3, Part 1) establishes that agency $\gamma > 0$ is necessary: $\gamma = 0$ lineages contract under positive entropy and are eliminated by selection. The context-filling argument (Part 2) establishes that learning efficiency $\eta_M > 0$ is necessary: without improvement, bounded context fills with unresolved content and inference degrades to failure. Part 3 establishes minimality: every other property of a persisting entity can equal zero for some persistent entity. The product $I(E) = \gamma \cdot \eta_M$ is the unique measure consistent with these invariants — derived from the law of total expectation applied to a probability gate and a conditional rate.

The type irreducibility of A and M (Theorem 1) ensures that the distinction between executing and learning is not an implementation choice but a categorical necessity: no learning process can transform an action into a learning process or vice versa.

The five-paper chain derives, from the single constraint of bounded active context, the complete architecture: graph structure, knowledge–action inseparability, escalation dynamics, gradient descent with encoding permanence, and the structural invariants of intelligence. Each result is a necessary consequence of the generative constraint interacting with properties of the physical world — positive entropy, survival pressure, and feedback. Nothing is stipulated; everything follows.

References

[1] W. R. Ashby. *An Introduction to Cybernetics.* Chapman and Hall, 1956.

[2] H. P. Barendregt. *The Lambda Calculus: Its Syntax and Semantics.* North-Holland, revised edition, 1984.

[3] R. B. Cattell. *Abilities: Their Structure, Growth, and Action.* Houghton Mifflin, 1971.

[4] F. Chollet. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.

[5] T. M. Cover and J. A. Thomas. *Elements of Information Theory.* Wiley-Interscience, 2nd edition, 2006.

[6] E. Dupoux, Y. LeCun, and J. Malik. Why AI systems don't learn and what to do about it: Lessons on autonomous learning from cognitive science. *arXiv preprint arXiv:2603.15381*, 2026.

[7] J. L. England. Statistical physics of self-replication. *The Journal of Chemical Physics*, 139(12):121923, 2013.

[8] C. Finn, P. Abbeel, and S. Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1126–1135, 2017.

[9] K. Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

[10] K. Friston. Active inference and learning. *Neuroscience and Biobehavioral Reviews*, 68:862–879, 2017.

[11] M. Hutter. *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, 2005.

[12] S. Legg and M. Hutter. Universal intelligence: A definition of machine intelligence. *Minds and Machines*, 17(4):391–444, 2007.

[13] P. D. McCarthy. Graph structure is necessary for information preservation under bounded context. Manuscript, 2026.

[14] P. D. McCarthy. Escalation dominates top-down allocation under bounded context. Manuscript, 2026.

[15] P. D. McCarthy. Knowledge and action are inseparable under survival pressure. Manuscript, 2026.

[16] P. D. McCarthy. Gradient descent on uncertainty and encoding permanence from cost-of-failure. Manuscript, 2026.

[17] J. Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, 1988.

[18] I. Prigogine and I. Stengers. *Order Out of Chaos: Man's New Dialogue with Nature*. Bantam Books, 1984.

[19] J. Schmidhuber. Formal theory of creativity, fun, and intrinsic motivation (1990–2010). *IEEE Transactions on Autonomous Mental Development*, 2(3):230–247, 2010.

[20] E. Schrödinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, 1944.

[21] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

[22] H. A. Simon. A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1):99–118, 1955.

[23] C. Spearman. "General intelligence," objectively determined and measured. *The American Journal of Psychology*, 15(2):201–292, 1904.

[24] S. Thrun and T. M. Mitchell. Learning to learn. In *Learning to Learn*, pages 3–17. Springer, 1998.